

063-0605-00L : Computational Structural Design 1

# Computational Graphic Statics

Dr. Juney Lee & Dr. Lluís Enrique

Autumn Semester 2021

## Week 2 Friday, October 1st

Hour 1

**9:45 – 10:30**

- Recap of Exercise 1
- **Lecture 2: Algorithmic design & thinking**

**10:30 – 10:45**

- 15 min break

Hour 2

**10:45 – 11:30**

- **Tutorial 2: Single node bridge in GH**

**11:30 – 11:45**

- 15 min break

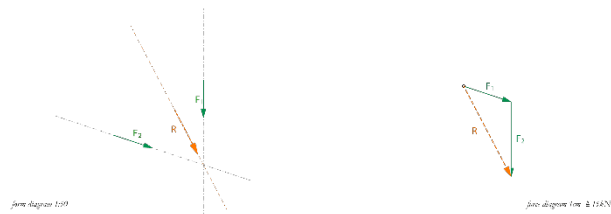
Hour 3

**11:45 – 12:30**

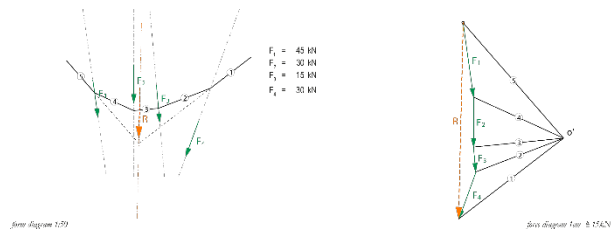
- **Tutorial 2: Single node bridge in GH (cont'd)**

**Task 1 Resultant of two non-parallel forces**

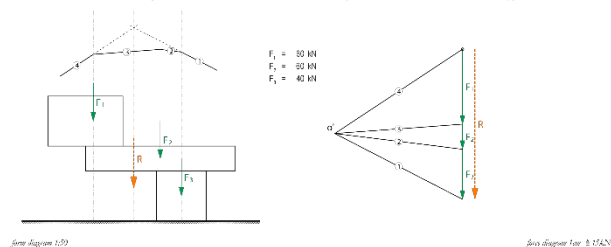
Find the position, direction and magnitude of the resultant in the form and force diagram with the help of vector addition.

**Task 2 Resultant of several non-parallel forces**

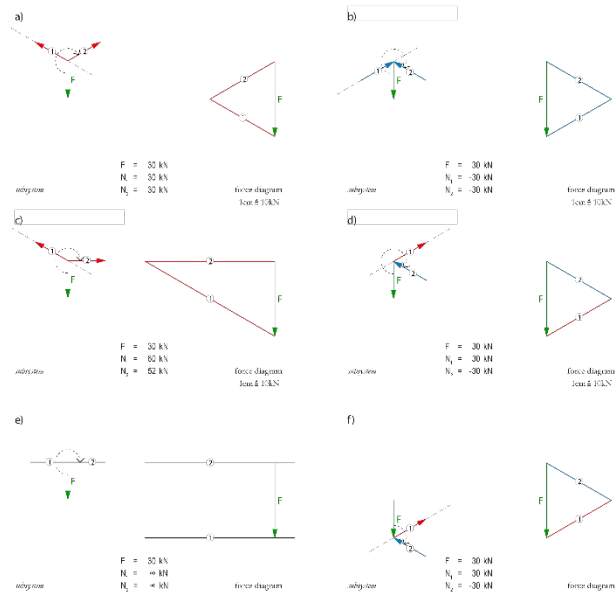
Find the position, direction and magnitude of the resultant in the form and force diagram with the help of a and funicular polygons.

**Task 3 Resultant of several parallel forces**

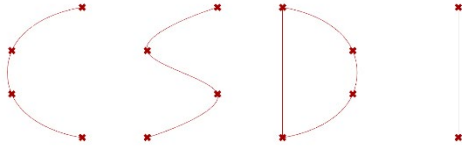
Find the position, direction and magnitude of the resultant in the form and force diagram with the help of a and funicular polygons.

**Task 4 Drawing subsystems**

Draw a corresponding force diagram for each subsystem (a-f). Determine the magnitude (kN) of each force on a marked direction in the subsystem. Indicate tension forces with red and compression forces with blue. Explain the solution of situation (e).



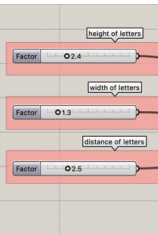
### 1. Point + line/curve construction



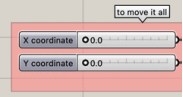
### 2. Coloured balloons



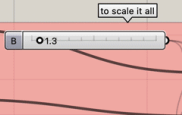
### 3. Height, width, spacing



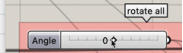
### 4. Position



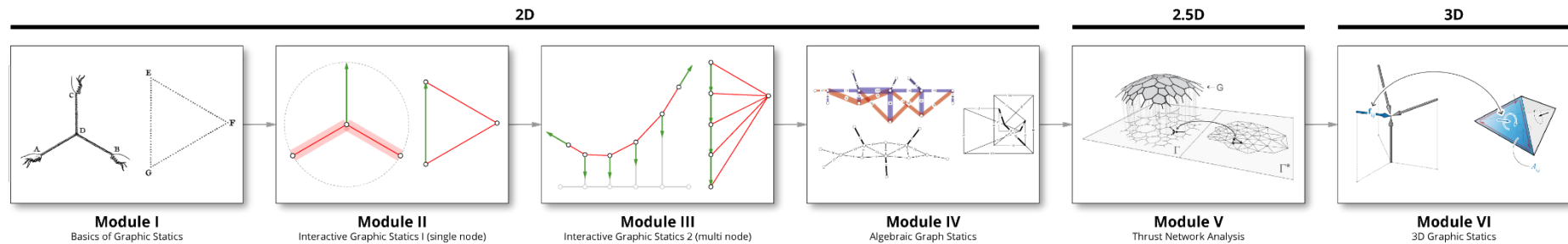
### 5. Scale

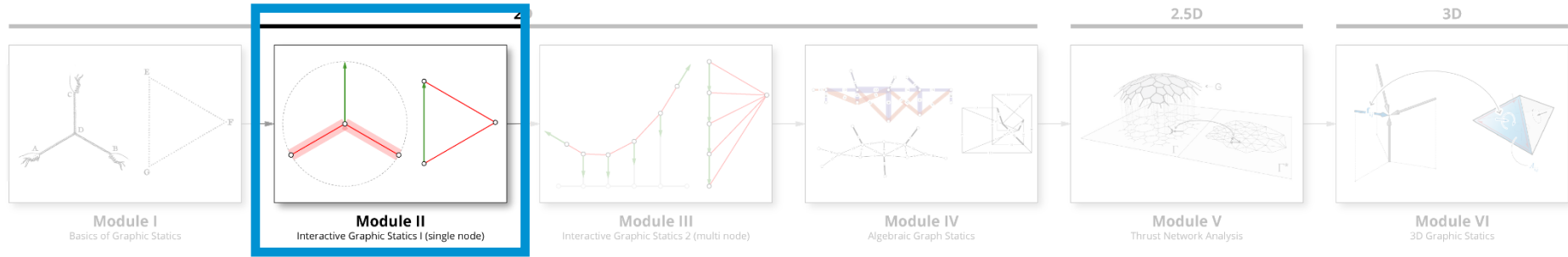


### 6. Rotation

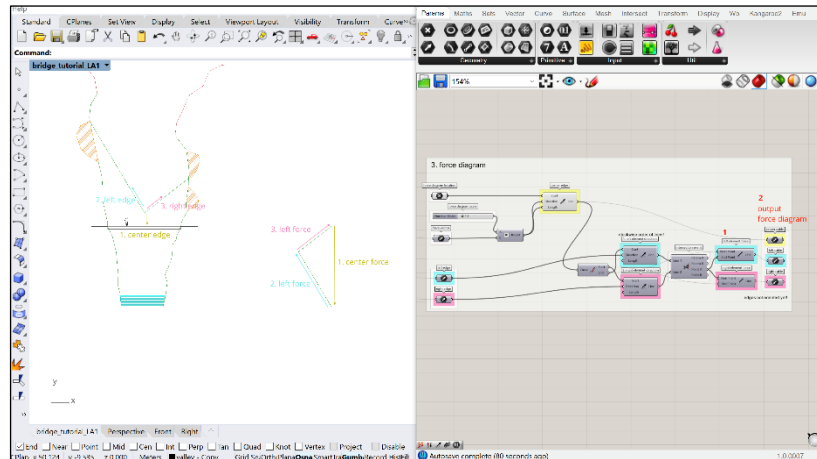


| Structural Typology   |                       | Topic              | Module           | Week               | hr                                                     | Topic                                    | Instructor                                           |                                       |                                 |  |
|-----------------------|-----------------------|--------------------|------------------|--------------------|--------------------------------------------------------|------------------------------------------|------------------------------------------------------|---------------------------------------|---------------------------------|--|
| 1. planar structures  | cables & arches       | Introduction       | I                | Week 1<br>(9/24)   | 1                                                      | Lecture 1                                | Introduction to graphic statics                      | Dr. Juney Lee                         |                                 |  |
|                       |                       |                    |                  |                    | 2                                                      | Tutorial 1                               | Basics of graphic statics on "blackboard"            | Dr. Lluis Enrique                     |                                 |  |
|                       |                       |                    |                  |                    | 3                                                      | Work session 1                           | Exercise 1: Reciprocal diagrams in Rhino + GH basics | Lotte Aldinger                        |                                 |  |
|                       |                       | 2D GS (procedural) | II               | Week 2<br>(10/1)   | 1                                                      | Lecture 2                                | Quick recap of Exercise 1                            | Lotte Aldinger                        |                                 |  |
|                       |                       |                    |                  |                    | 2                                                      | Tutorial 2                               | Algorithmic design & thinking                        | Dr. Juney Lee                         |                                 |  |
|                       |                       |                    |                  |                    | 3                                                      |                                          | Single node bridge in GH (form to force)             | Lotte Aldinger                        |                                 |  |
|                       | III                   |                    | Week 3<br>(10/8) | 2                  |                                                        |                                          |                                                      |                                       |                                 |  |
|                       |                       |                    |                  | 3                  |                                                        |                                          |                                                      |                                       |                                 |  |
|                       |                       |                    |                  | Week 4<br>(10/15)  | 1                                                      | Lecture 3                                | Quick recap of Exercise 2                            | Lotte Aldinger                        |                                 |  |
|                       |                       | 2                  | Tutorial 3       |                    | Computational graphic statics                          | Dr. Juney Lee                            |                                                      |                                       |                                 |  |
|                       |                       | 3                  |                  |                    | Multi-node bridge in Grasshopper                       | Dr. Lluis Enrique                        |                                                      |                                       |                                 |  |
|                       |                       | trusses            | 2D GS (AGS)      | IV                 | Week 5<br>(10/22)                                      | 1                                        | Work session 3                                       | Exercise 3: Multi-node bridge with GH | Dr. Lluis Enrique + instructors |  |
|                       | 2                     |                    |                  |                    |                                                        |                                          |                                                      |                                       |                                 |  |
|                       | 3                     |                    |                  |                    |                                                        |                                          |                                                      |                                       |                                 |  |
|                       |                       |                    |                  | Week 6             | seminar week                                           |                                          |                                                      |                                       |                                 |  |
|                       | 2.5 GS (TNA)          |                    | V                | Week 7<br>(11/5)   | 1                                                      | Lecture 4                                | Quick recap of Exercise 3                            | Dr. Lluis Enrique                     |                                 |  |
| 2                     |                       |                    |                  |                    | Tutorial 4                                             | Algebraic graph statics (AGS)            | Dr. Juney Lee                                        |                                       |                                 |  |
| 3                     |                       |                    |                  |                    | compas_agss + IGS (Interactive Graphic Statics) plugin | Ricardo Avelino                          |                                                      |                                       |                                 |  |
| 3D GS (polyhedral)    | VI                    | Week 8<br>(11/12)  | 1                | Work session 4     | Exercise 4: Truss problems with IGS                    | Ricardo Avelino + instructors            |                                                      |                                       |                                 |  |
|                       |                       |                    | 2                |                    |                                                        |                                          |                                                      |                                       |                                 |  |
|                       |                       |                    | 3                |                    |                                                        |                                          |                                                      |                                       |                                 |  |
| 2. surface structures | shells                | 2.5 GS (TNA)       | V                | Week 9<br>(11/19)  | 1                                                      | Lecture 5                                | Quick recap of Exercise 4                            | Ricardo Avelino                       |                                 |  |
|                       |                       |                    |                  |                    | 2                                                      | Tutorial 5                               | Thrust Network Analysis (TNA)                        | Dr. Juney Lee                         |                                 |  |
|                       |                       |                    |                  |                    | 3                                                      |                                          | compas_tna + RV2 (RhinoVAULT 2) plugin               | Dr. Juney Lee                         |                                 |  |
|                       | polyhedral structures | 3D GS (polyhedral) | VI               | Week 10<br>(11/26) | 1                                                      | Work session 5                           | Exercise 5: Shell design exercises with RV2          | Dr. Juney Lee + instructors           |                                 |  |
|                       |                       |                    |                  |                    | 2                                                      |                                          |                                                      |                                       |                                 |  |
|                       |                       |                    |                  |                    | 3                                                      |                                          |                                                      |                                       |                                 |  |
| 3. spatial structures | polyhedral structures | 3D GS (polyhedral) | VI               | Week 11<br>(12/3)  | 1                                                      | Lecture 6                                | Quick recap of Exercise 5                            | Dr. Juney Lee                         |                                 |  |
|                       |                       |                    |                  |                    | 2                                                      | Tutorial 6                               | 3D graphic statics (3GS)                             | Dr. Juney Lee                         |                                 |  |
|                       |                       |                    |                  |                    | 3                                                      | Work session 6                           | compas_3gs + 3GS (3D graphic statics) plugin         | Dr. Juney Lee                         |                                 |  |
|                       | polyhedral structures | 3D GS (polyhedral) | VI               | Week 12<br>(12/10) | 1                                                      | Lecture 7                                | Exercise 6: Spatial structures with 3GS              | Dr. Juney Lee                         |                                 |  |
|                       |                       |                    |                  |                    | 2                                                      |                                          | Quick recap of Exercise 6                            | Dr. Juney Lee                         |                                 |  |
|                       |                       |                    |                  |                    | 3                                                      |                                          | Pt I. Outlook on computational graphic statics       | Dr. Juney Lee                         |                                 |  |
|                       |                       |                    |                  |                    |                                                        | Pt II. Guest Lecture                     | TBD                                                  |                                       |                                 |  |
|                       |                       |                    |                  |                    |                                                        | Discussions, feedback, evaluations, etc. |                                                      |                                       |                                 |  |





## Module II · 2D procedural graphic statics 1



+



063-0605-00L : Computational Structural Design 1  
Computational Graphic Statics

# Lecture 2

## Algorithmic design & thinking

Friday, October 1st, 2021

Dr. Juney Lee

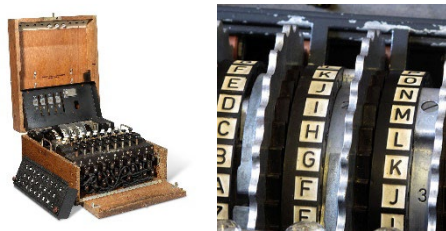
# Computational Graphic Statics

## Universal Computing "Turing" Machine

... a machine capable of computing anything that is computable by executing instructions (program) stored on tape (memory), allowing the machine to be programmable ...

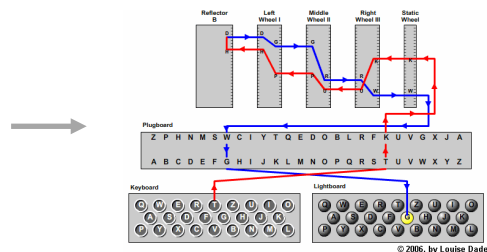
"On Computable Numbers" | Alan Turing (1936)

ATTACKATDAWN



Encryption machine ("Enigma")

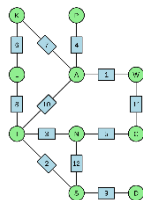
"Scrambler" using rotor configurations



Encryption key

Algorithm

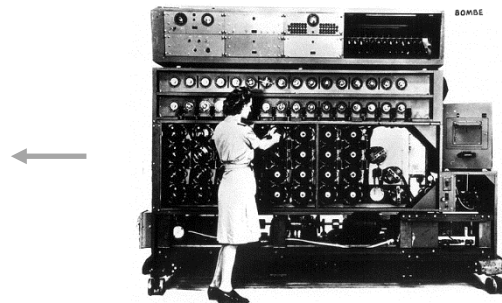
WSNPNLKLSTCS



|                            |   |   |   |   |   |   |   |   |   |    |    |    |
|----------------------------|---|---|---|---|---|---|---|---|---|----|----|----|
| <b>Ciphertext</b>          | W | S | N | P | N | L | K | L | S | T  | C  | S  |
| <b>Plaintext "crib"</b>    | A | T | T | A | C | K | A | T | D | A  | W  | N  |
| <b>Message position</b>    | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
| <b>Upper drum setting</b>  | Z | Z | Z | Z | Z | Z | Z | Z | Z | Z  | Z  | Z  |
| <b>Middle drum setting</b> | Z | Z | Z | Z | Z | Z | Z | Z | Z | Z  | Z  | Z  |
| <b>Lower drum setting</b>  | A | B | C | D | E | F | G | H | I | J  | K  | L  |

Decryption key

Reverse-algorithm



Decryption machine

Multiple reverse-engineered Enigma machines

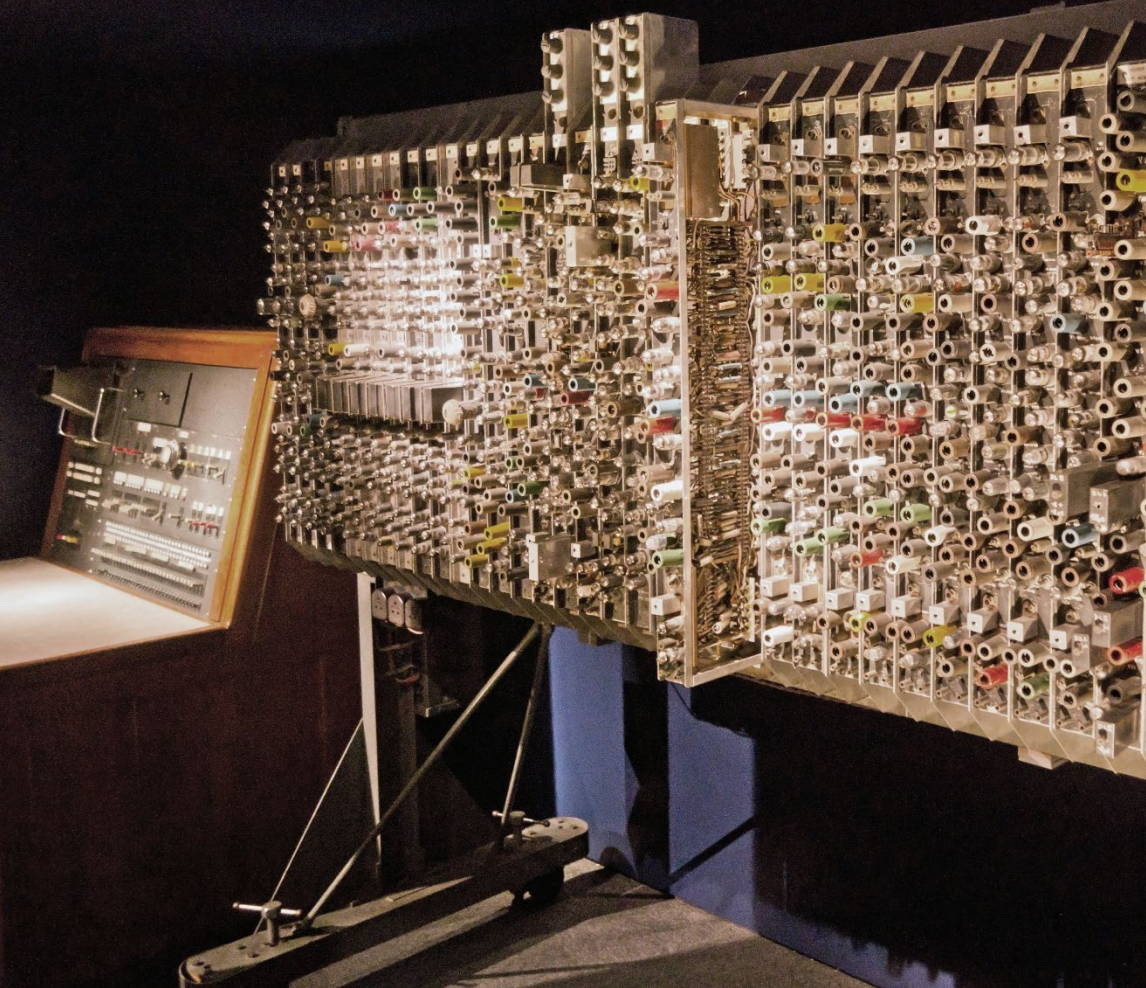


Photo: Antoine Taveneaux

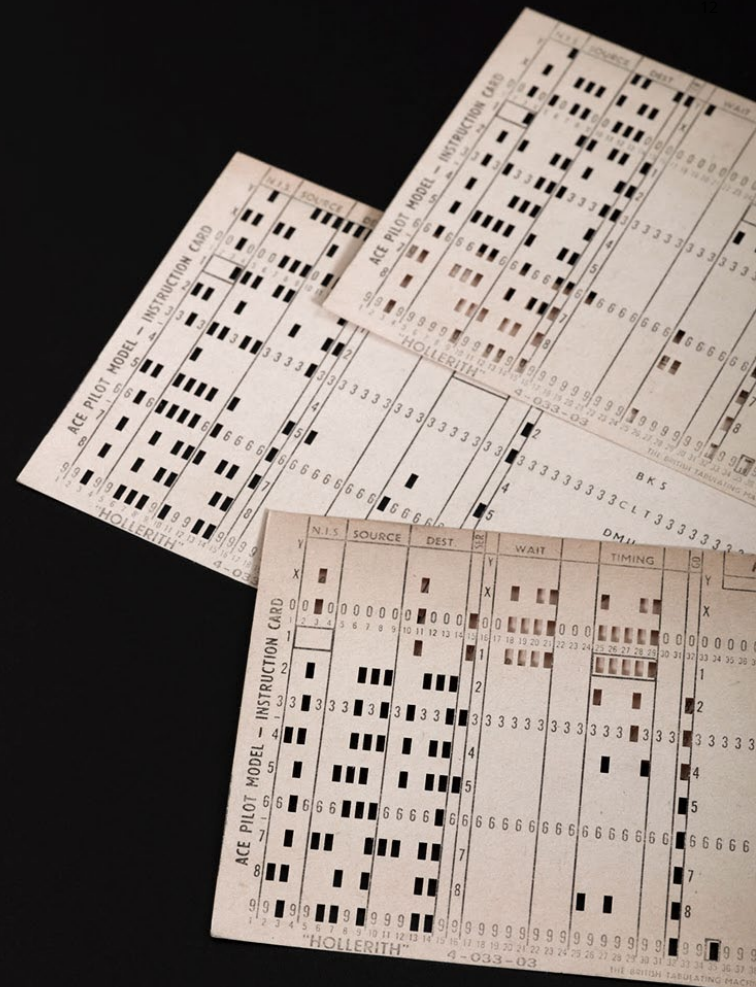


Photo: Science Museum London

# SKETCHPAD

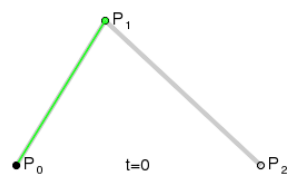


TX-2 computer

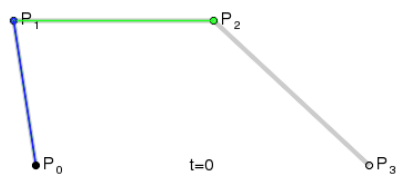
Industrialising CAD



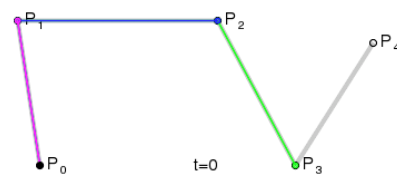
Boeing factory



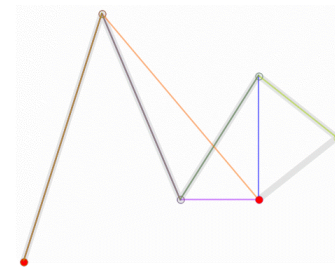
quadratic Bézier curve



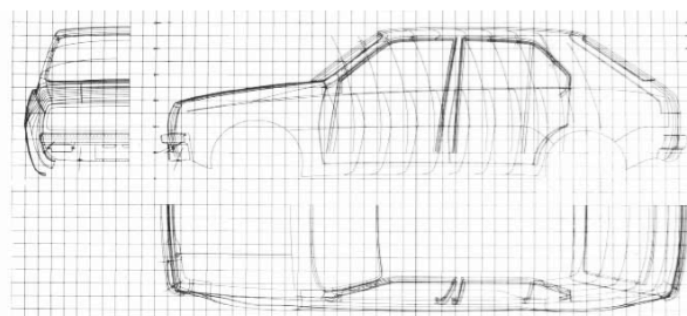
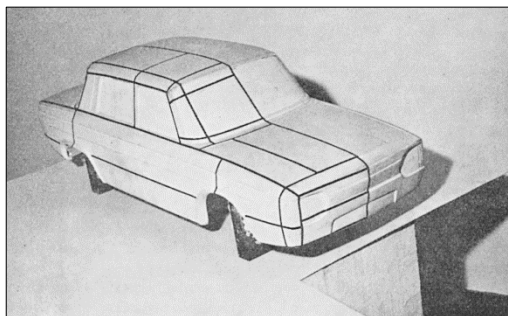
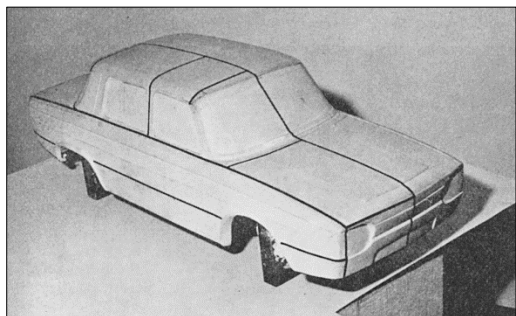
quadratic Bézier curve



cubic Bézier curve



fifth-order Bézier curve





### **Xerox Alto (1979)**

First "desktop" with graphical user interface (GUI)  
 US\$32,000 in 1979 (equivalent to US\$112,726)



### **Epson HX-20 (1982)**

first "true" laptop computer  
 US\$795 (equivalent to US\$2040.76)



### **Apple Lisa (1983)**

Individual computing with GUI  
 US\$9,995 (equivalent to US\$25,811.18)



### **Macintosh (1984)**

First "personal computer"  
 US\$2,495 (equivalent to US\$6,100)

# Computer-aided Design

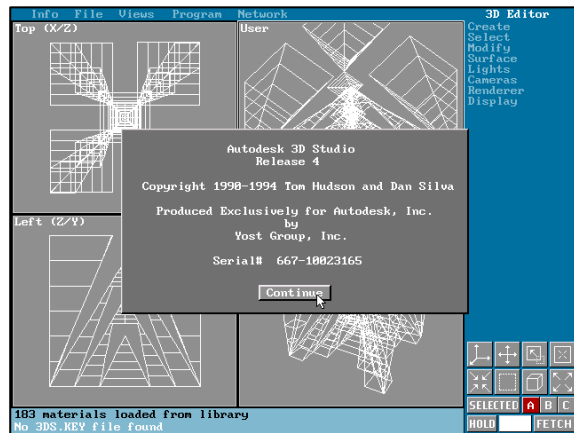


AutoCAD v1.0 (1982)

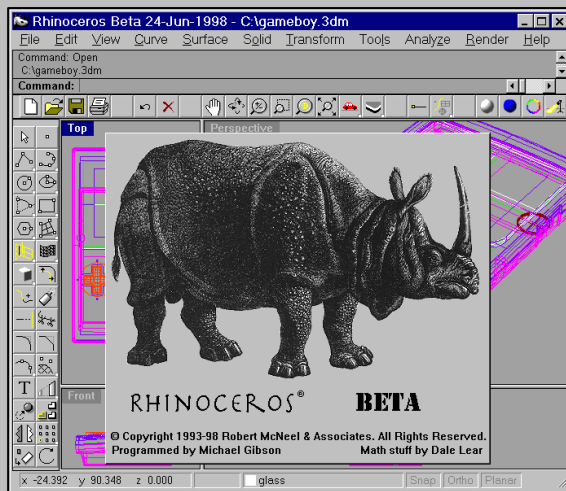


CATIA 1 (1982)

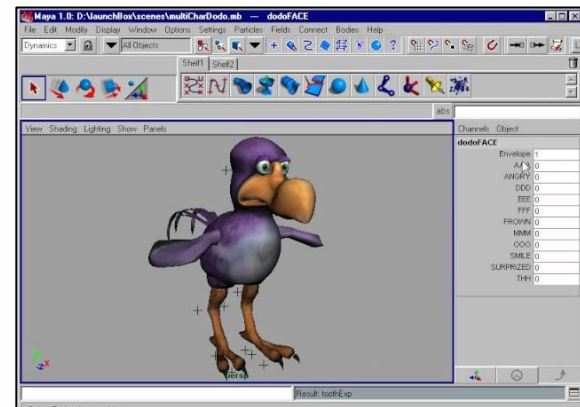




3D Studio Max (1990)



Rhinoceros/Sculptura 2 (1993)



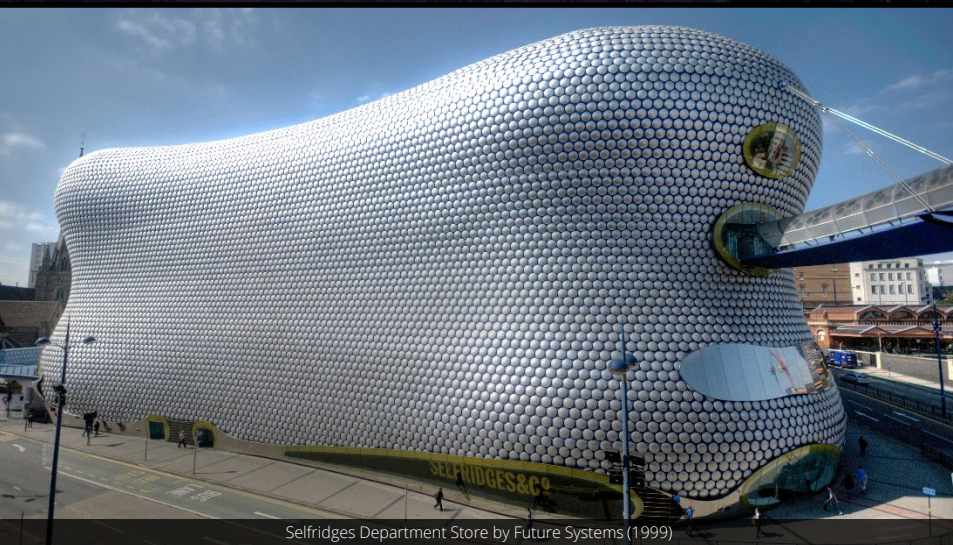
Maya (1998)



Sage Gateshead by Foster + Partners (2004)



Kunsthhaus Graz by Colin Fournier (2003)



Selfridges Department Store by Future Systems (1999)

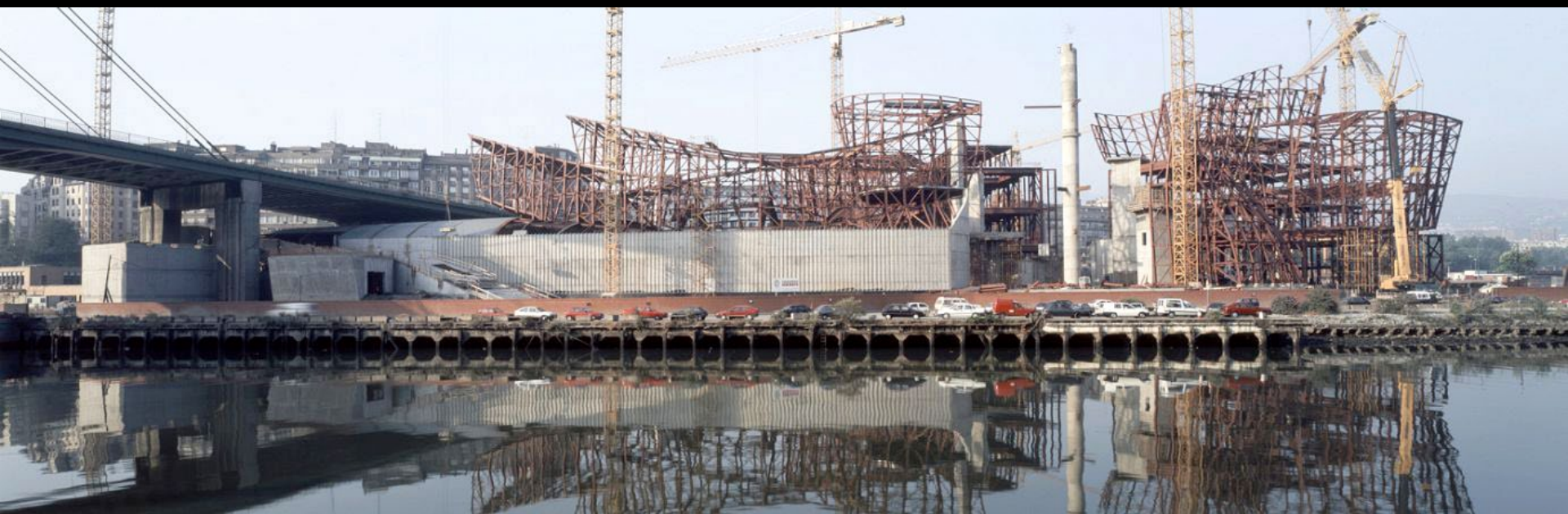


Guggenheim Museum Bilbao by Frank Gehry (1997)

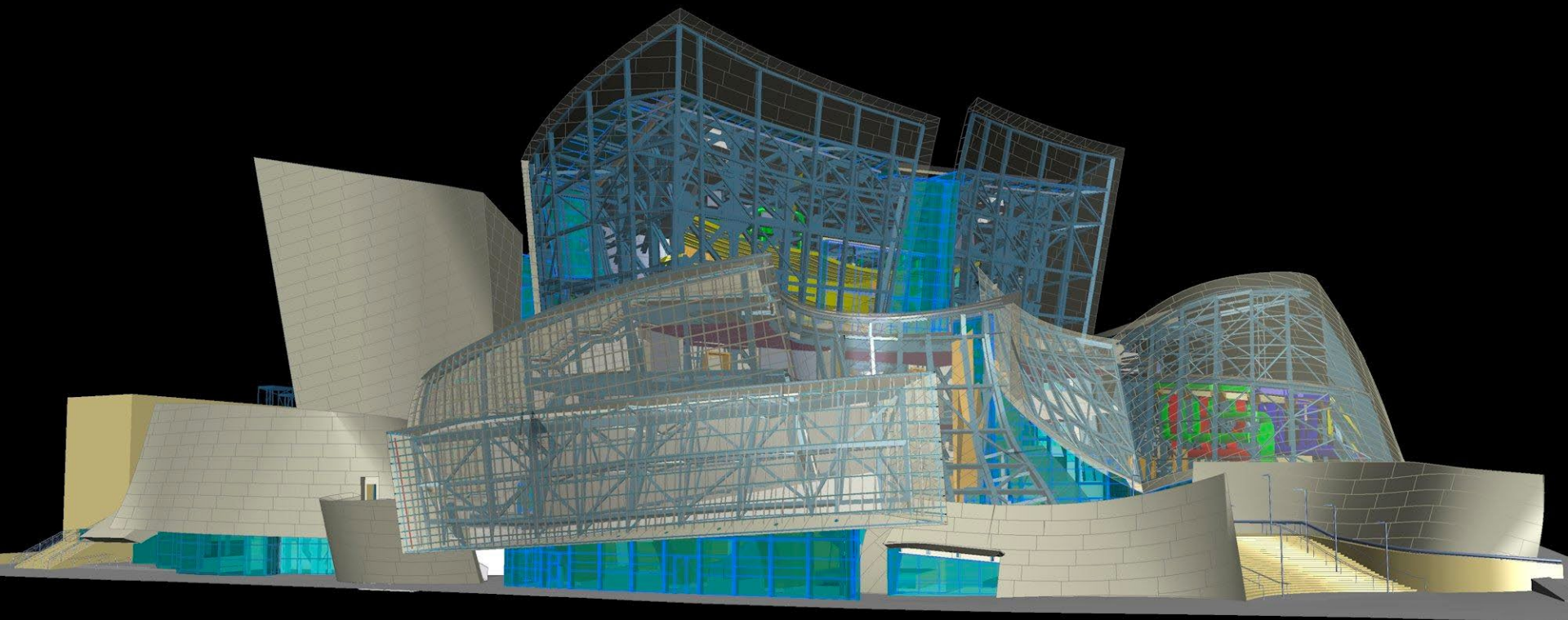


← Frank Gehry

emio

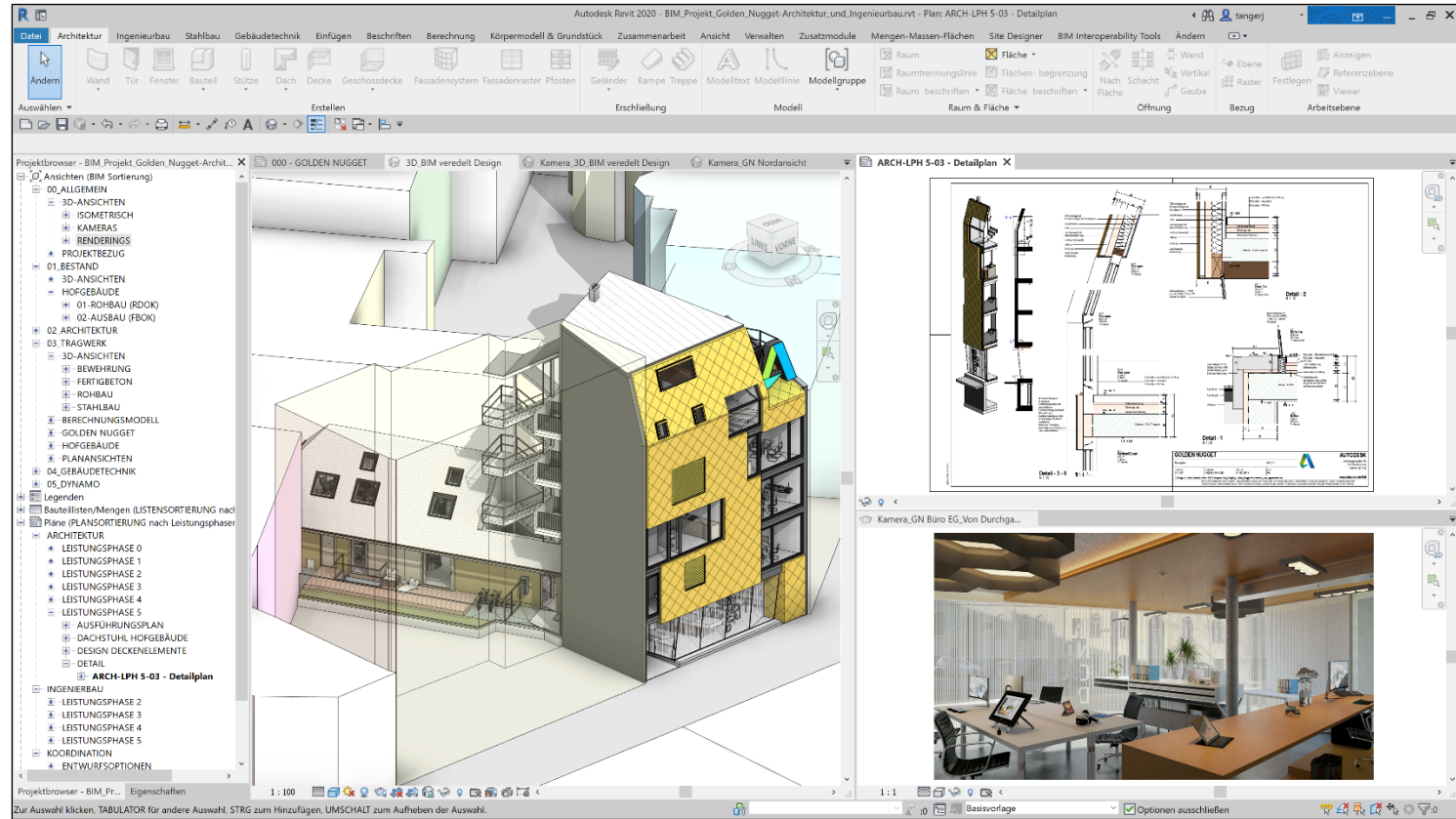


Construction of Guggenheim Museum, Bilbao, Spain, 1997 | Frank Gehry

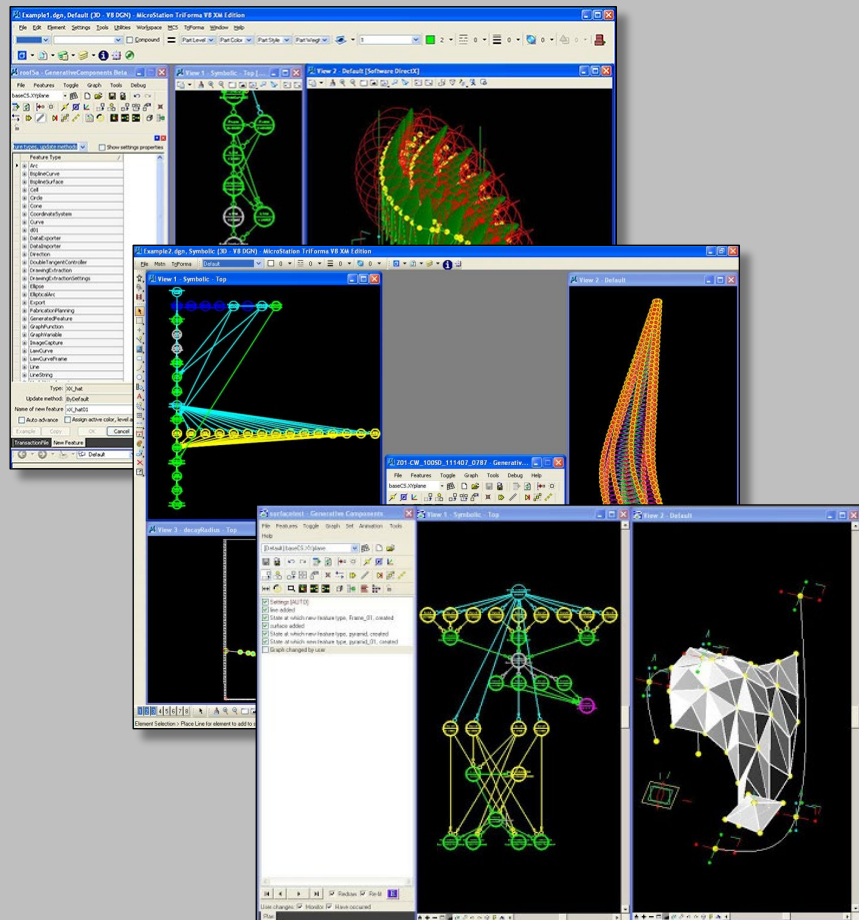


Walt Disney Concert Hall (Digital Project 3D model), Los Angeles, USA, 2003, Frank Gehry

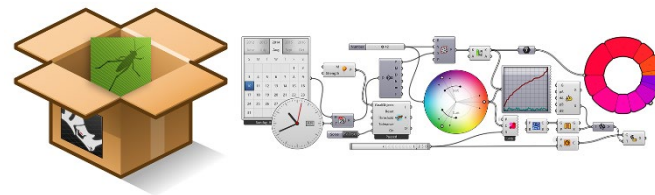
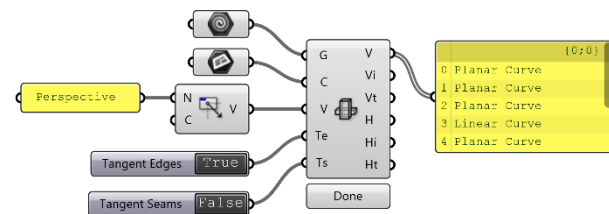
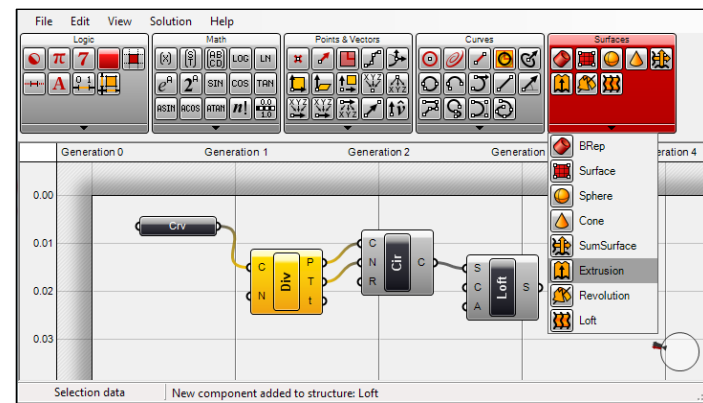




# Computational ~~Computer-aided~~ Design



Generative Design (2003)



Explicit History / Grasshopper (2007)



Metropol Parasol by Jürgen Mayer



Galaxy SOHO by Zaha Hadid Architects

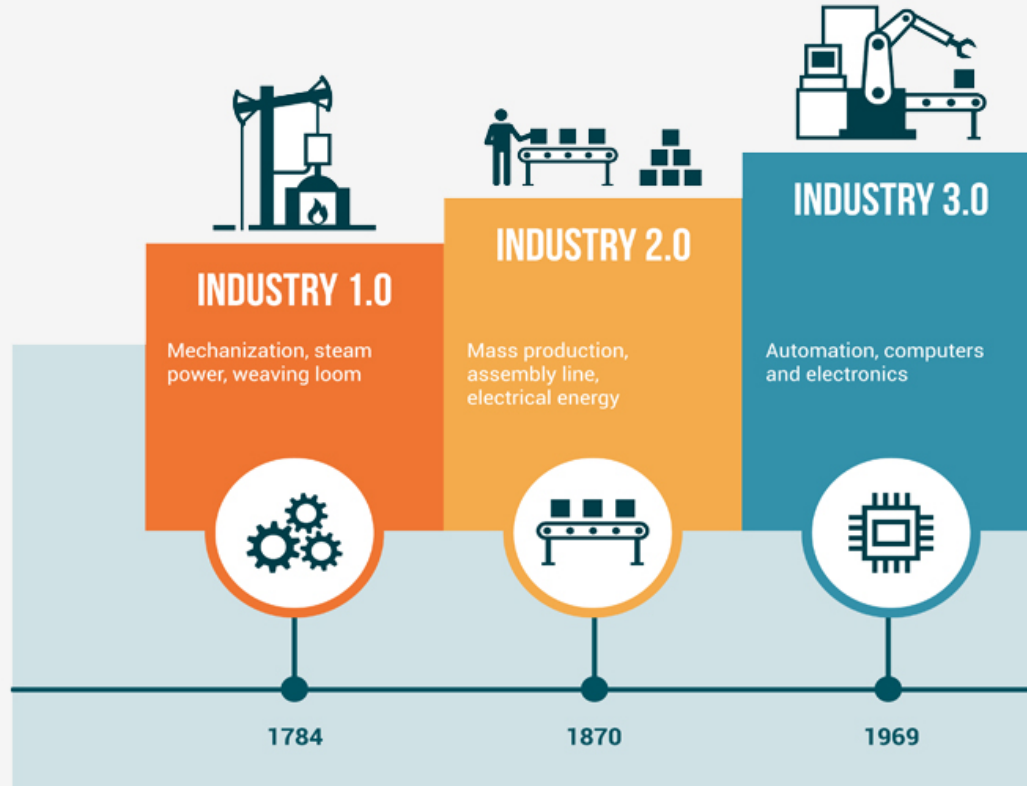


Serpentine Gallery Pavillion by BIG



Pillar of Dreams by Marc Fornes

Moore's law describes the empirical regularity that the number of transistors on integrated circuits doubles approximately every two years. This advancement is important as other aspects of technological progress – such as processing speed or the price of electronic products – are linked to Moore's law.

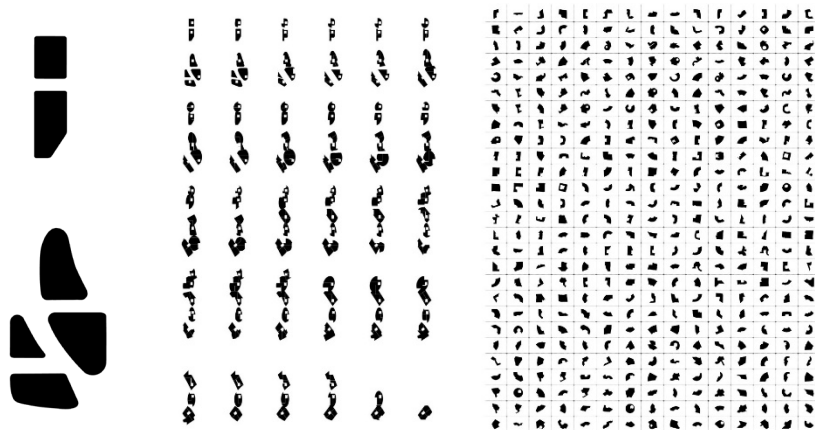




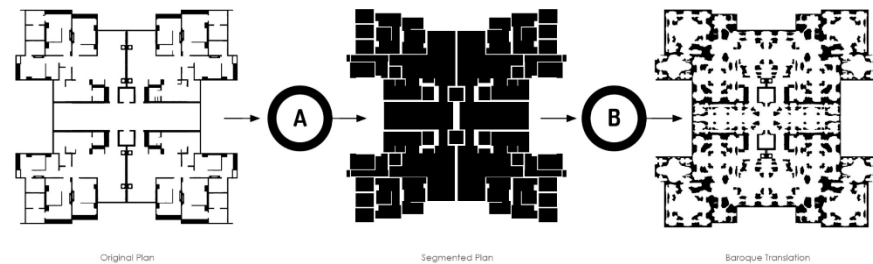
Layout + spatial configuration algorithms



Generative design + topology optimisation

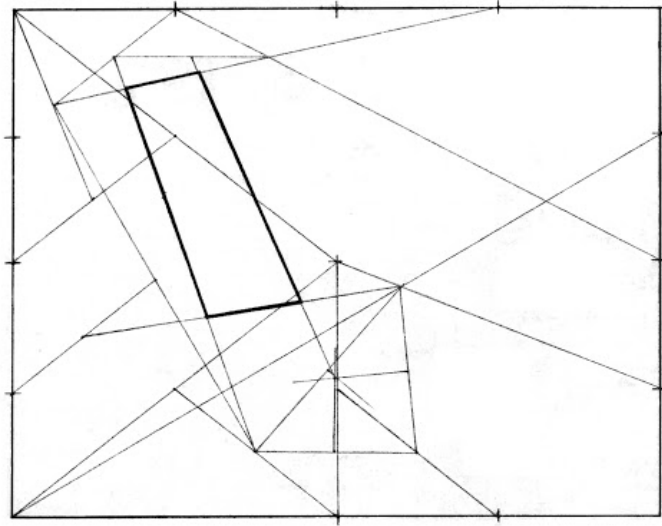


Pattern recognition + cataloguing



Style crossover

# Computational Design Algorithmic

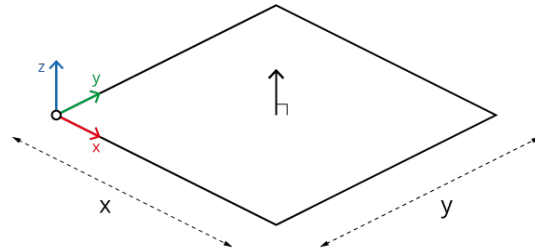


A quadrangle which is formed and enclosed by four lines, the first of which is drawn from a point halfway between a point halfway between the center of the wall and the upper left corner and the midpoint of the left side and the upper left corner to a point halfway between the midpoint of the top side and the upper right corner, the second line from a point halfway between the start of the first line and a point halfway between the midpoint of the top side and the upper left corner to a point halfway between a point halfway between the center of the wall and the lower left corner and the midpoint of the bottom side, the third line from a point halfway between a point halfway between the start of the first line and the end of the second line and a point halfway between the midpoint of the left side and the lower left corner to a point which is on an axis between the lower left corner to a point halfway between the midpoint of the right side and the upper right corner where a line drawn from the center of the wall to a point halfway between the midpoint of the right side and the lower right corner would cross that axis, the fourth line from a point equidistant from the end of the third line, the end of the second line and a point halfway between a point halfway between the center of the wall and the midpoint of the bottom side and a point halfway between the midpoint of the bottom side and the lower right corner to a point halfway between the start of the second line and a point where a line would cross the first line if it were drawn from the midpoint of the right side to a point halfway between the midpoint of the top side and the upper left corner.

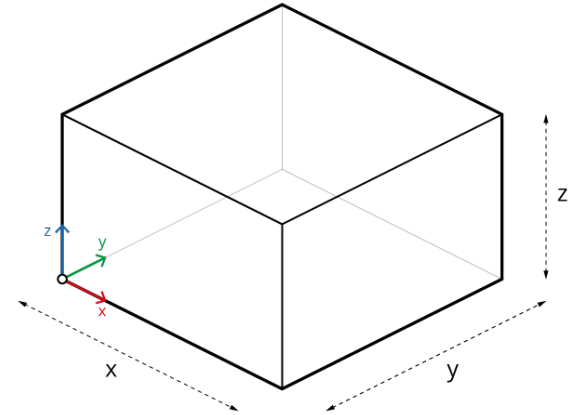
—Sol LeWitt, 1974



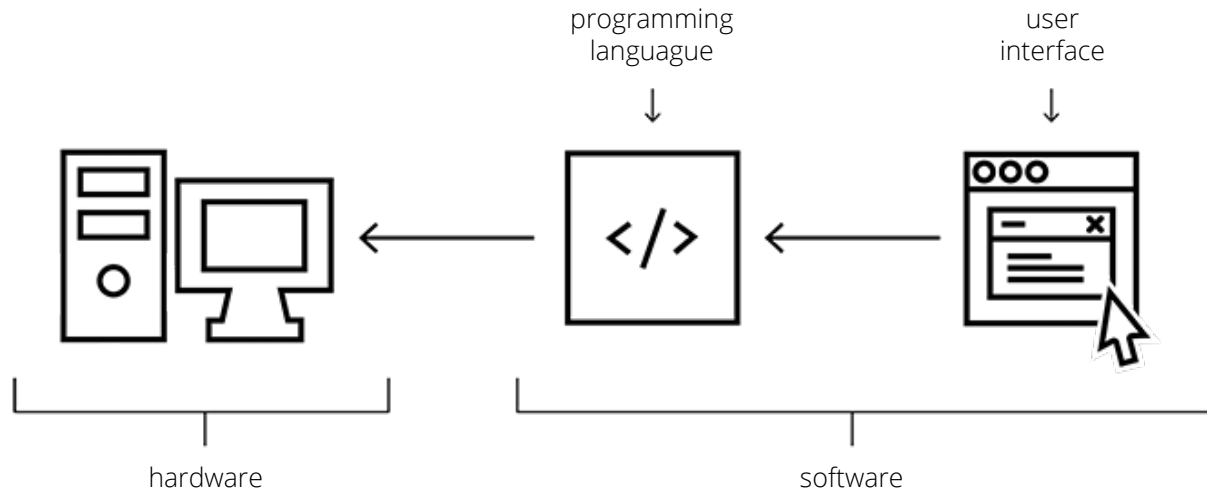
1. Choose a starting point

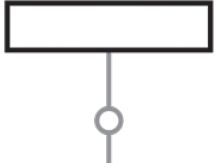


2. Draw a rectangle with dimensions X and Y on the xy plane

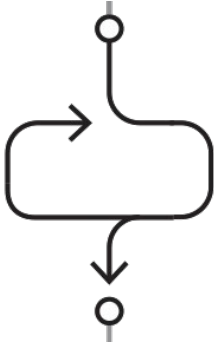


3. Extrude the rectangle in the z-axis with amount z

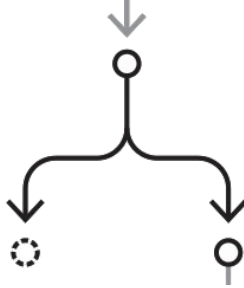




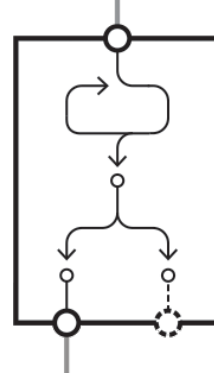
1. variables



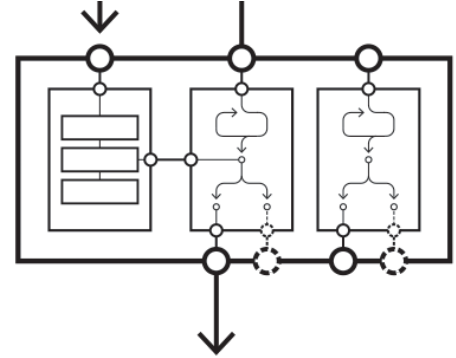
2. loops



3. conditional

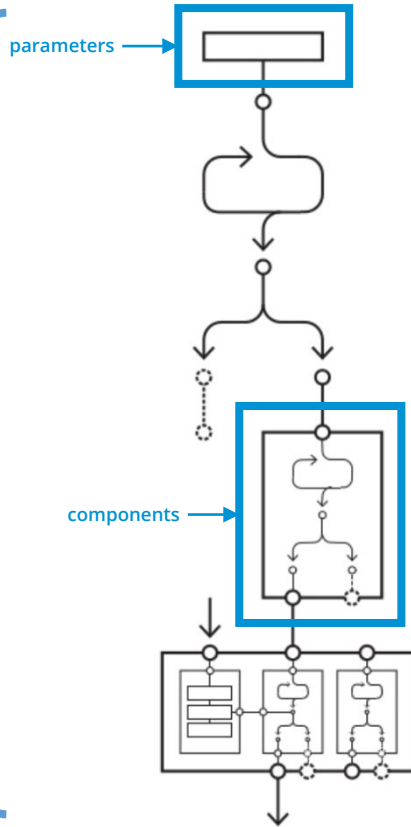


4. functions



5. objects

visual  
programming  
(Grasshopper)



1. variable(s)

```
x = 1
y = 'hello world'
myList = [1, 'b', 3]
myDict = {'a': 1, 'b': 2}
```

2. Loops

```
for i in range(10):
    # do something 10 times

for element in myList:
    # do something for all elements
```

3. Conditional

```
if x > 10 and y > 10:
    # do something
elif x > 5 or y < 5:
    # do something else
else:
    # do something else
```

4. Functions

```
def myFunction(input1, input2):
    # do something
    return output

c = myFunction(a, b)
```

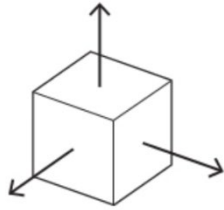
5. Object

```
class MyClass:
    def __init__(self, input1):
        self.localVar = input1
    def myMethod(self, input2):
        # do something
        return output

classInstance = MyClass(a)
c = classInstance.myMethod(b)
```

procedural  
programming  
("scripting")

object-oriented  
programming  
("OOP")

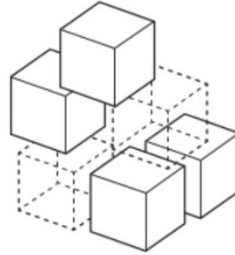


### Morphological

- Continuous measures

+ Good top-down control over design

- Can usually only generate simple design spaces

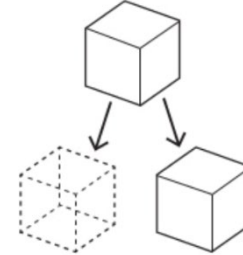


### State-change

- Choices, categories

+ can create discontinuous design spaces  
+ control over individual elements

- Many inputs required (each element needs to be controlled separately)

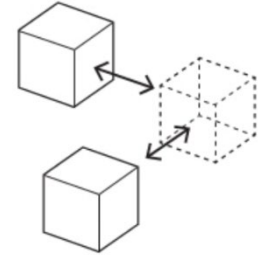


### Rule-based

- L-system, shape grammars, 1D cellular automata

+ reduced number of inputs (abstraction of inputs into rule sets)  
+ can create complexity

- Only top-down control
- can't control individual behaviour
- can't create emergence
- Potentially redundant or incomplete design space

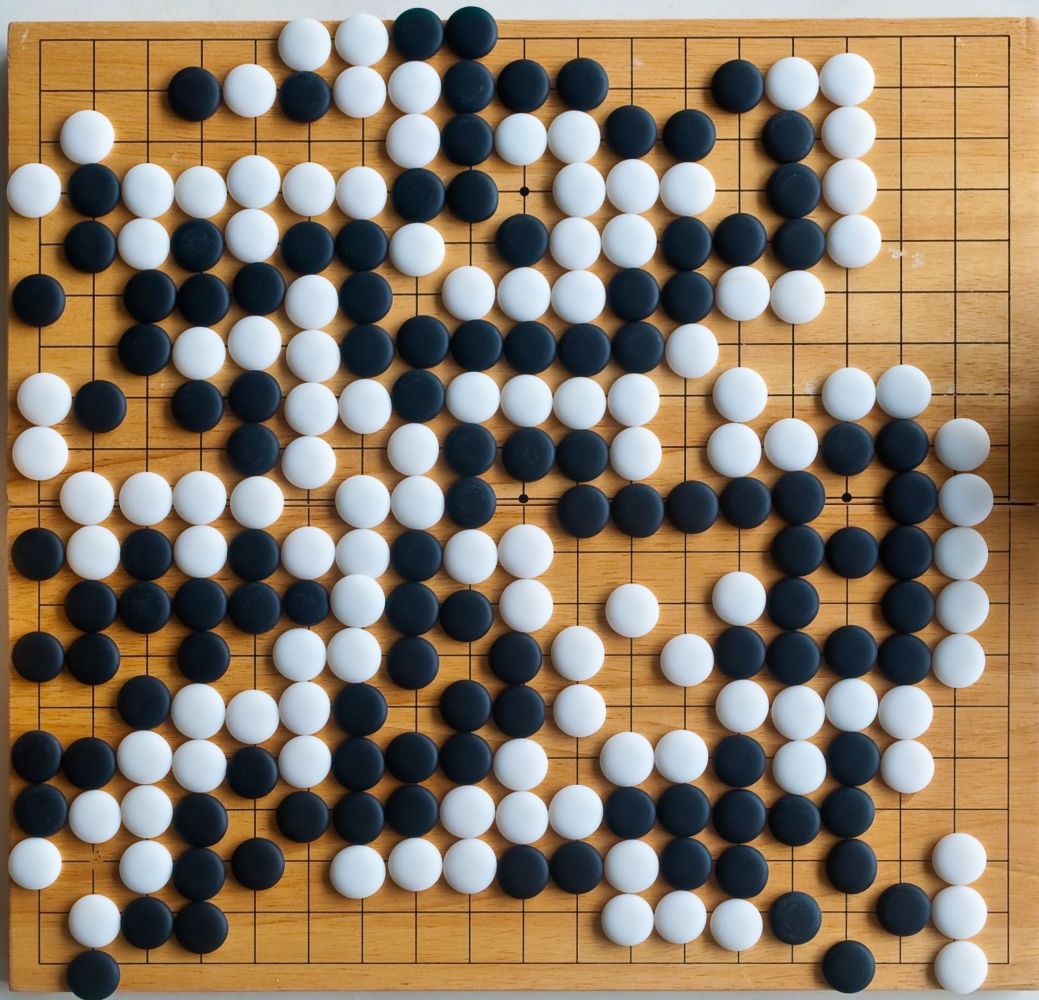


### Behavioral

- object-oriented, agent-based behavior models (dynamic)

+ reduced number of inputs (abstraction of inputs into agent behaviors)  
+ can lead to emergence

- Little intuitive control over macro design
- Potentially redundant or incomplete design space



Overlay:

☐ None

PV

Prior

## Rollout

VNet

mixed

VIEWS



149



## Gigo

Rank

## Gigo

Rank

## Comments

10209 01:56:36.261

I0209 01:56:36.510

10209 01:56:36.875

I0209 01:56:36.875

I0209 01:56:36.875

I0209 01:56:36.88

10209 01:56:36.88

10209 01:56:36.88

I0209 01:56:36.88

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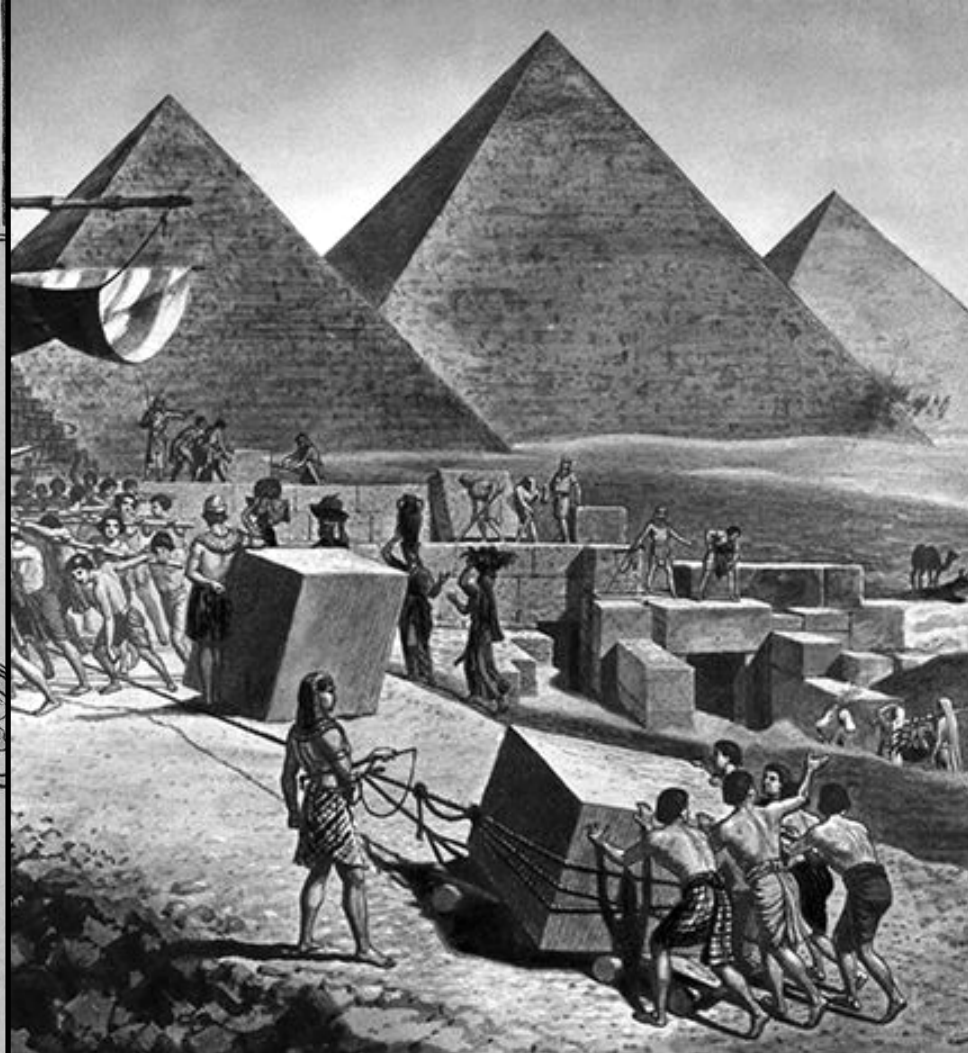
Move number to pl

Last move: g9

Last ko point: in

EDGE RO POINT, 41









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"The purpose of computing is **insight**,  
not numbers."

## Richard Hamming (1962)

063-0605-00L : Computational Structural Design 1  
Computational Graphic Statics

<http://block.arch.ethz.ch>

 @blockresearchgroup